

Scalability & Future Plan

Date	06 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited loan dataset	Prediction accuracy may decrease when tested on diverse real-world applicant data.	High
2	Limited applicant parameters	The system considers only predefined applicant features, which may not represent all real banking scenarios.	Medium
3	Static machine learning model	The model requires retraining whenever new loan data becomes available.	Medium
4	No user authentication	Anyone with access to the application can perform loan predictions without secure login.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited users on Render free tier	Upgrade to Render Starter/Standard plan or deploy on scalable cloud infrastructure such as AWS or Azure.
2	Data Storage	Uses local dataset and pickle model files	Integrate MySQL or MongoDB for secure storage of applicant records and prediction history.
3	Prediction Performance	Uses a trained Random Forest model with Flask backend.	Optimize prediction speed and retrain the model periodically using larger datasets and advanced ensemble techniques.
4	Security	Basic web application without authentication.	Implement user authentication, encrypted password storage, HTTPS, and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Improve prediction accuracy using larger loan datasets and advanced ensemble techniques such as XGBoost and LightGBM.	Short-term	Higher prediction accuracy and better generalization.
Phase 3	Integrate user authentication, prediction history, and an interactive dashboard with graphical analytics.	Mid-term	Improved security, better user experience, and enhanced data visualization.
Phase 4	Integrate real-time banking APIs, cloud databases, and automated model retraining for real-world deployment.	Long-term	Real-time loan prediction, increased scalability, and enterprise-level deployment.